

Module -2

Transient Analysis of First Order RC Circuits

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Introduction

- A first order circuit is characterized by a first order differential equation.
- Types of Circuits:
 1. RC circuit
 2. RL circuit
- Types of excitation:
 1. Source free excitation (Natural response)
 2. Independent source excitation (Forced response)

- Source free excitation:

- ❖ The energy is initially assumed to be stored in the inductive or capacitive element.
- ❖ This energy causes current to flow in the circuit and is gradually dissipated in the resistors.
- ❖ The source free circuits do not have independent sources, whereas dependent sources may exist.

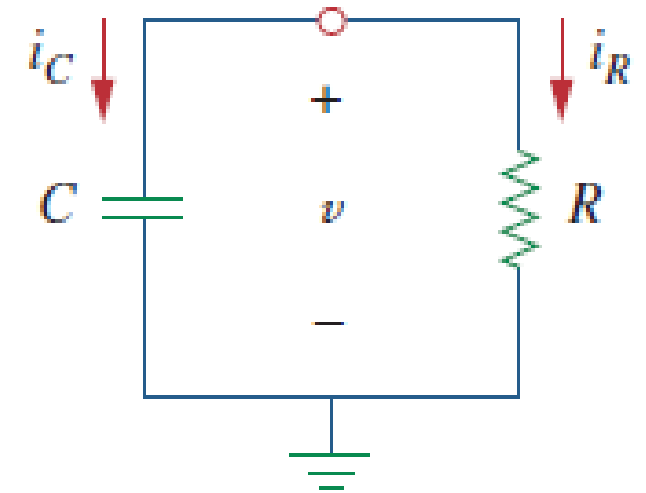
- Independent source excitation:

- ❖ Independent sources are used to excite the circuits.
- ❖ Normally DC sources are considered here.

2 types of circuits and 2 types of excitation = 4 possible situations.

The Source Free RC circuit

- A source free RC circuit occurs when its dc source is suddenly disconnected.
- The energy already stored in the capacitor is released to the resistors.
- C is the equivalent capacitance
- R is the equivalent resistance
- To determine the response of the circuit is the objective.



- Let $v(t)$ be the initial voltage across the capacitor. The initial voltage $v(0) = V_0$ at time $t=0$

$$v(0) = V_0$$

- The energy stored is

$$w(0) = \frac{1}{2}CV_0^2$$

Applying KCL at the top node of the circuit

$$i_C + i_R = 0$$

$$C \frac{dv}{dt} + \frac{v}{R} = 0$$

$$\frac{dv}{dt} + \frac{v}{RC} = 0$$

This is a *first-order differential equation*.

$$\frac{dv}{v} = -\frac{1}{RC} dt$$

Integrating both sides,

$$\ln v = -\frac{t}{RC} + \ln A$$

where $\ln A$ is the integration constant.

$$\ln \frac{v}{A} = -\frac{t}{RC}$$

Taking powers of e produces

$$v(t) = Ae^{-t/RC}$$

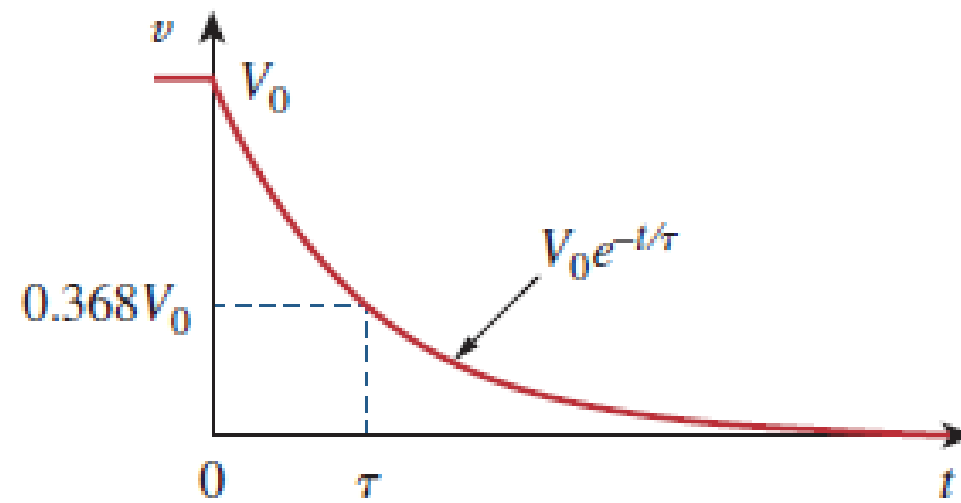
But from the initial conditions, $v(0) = A = V_0$.

$$v(t) = V_0 e^{-t/RC}$$

The voltage response of a RC circuit is an exponential decay response of the initial voltage and is called Natural response of the circuit.

The **natural response** of a circuit refers to the behavior (in terms of voltages and currents) of the circuit itself, with no external sources of excitation.

The voltage response of the RC circuit.



Values of $v(t)/V_0 = e^{-t/\tau}$.

t	$v(t)/V_0$
τ	0.36788
2τ	0.13534
3τ	0.04979
4τ	0.01832
5τ	0.00674

The **time constant** of a circuit is the time required for the response to decay to a factor of $1/e$ or 36.8 percent of its initial value.¹

$$V_0 e^{-\tau/RC} = V_0 e^{-1} = 0.368V_0$$

$$\tau = RC$$

$$v(t) = V_0 e^{-t/\tau}$$

$$i_R(t) = \frac{v(t)}{R} = \frac{V_0}{R} e^{-t/\tau}$$

The power dissipated in the resistor is

$$p(t) = vi_R = \frac{V_0^2}{R} e^{-2t/\tau}$$

The energy absorbed by the resistor up to time t is

$$\begin{aligned} w_R(t) &= \int_0^t p(\lambda) d\lambda = \int_0^t \frac{V_0^2}{R} e^{-2\lambda/\tau} d\lambda \\ &= -\frac{\tau V_0^2}{2R} e^{-2\lambda/\tau} \Big|_0^t = \frac{1}{2} C V_0^2 (1 - e^{-2t/\tau}), \quad \tau = RC \end{aligned}$$

Notice that as $t \rightarrow \infty$, $w_R(\infty) \rightarrow \frac{1}{2} C V_0^2$,

- **Steps to solve the Source free RC circuit:**

1. Find the initial voltage V_0 across the capacitor.
2. Find the time constant
3. Find the voltage across the capacitor.

$$v(t) = V_0 e^{-t/\tau}$$

Numerical - 1

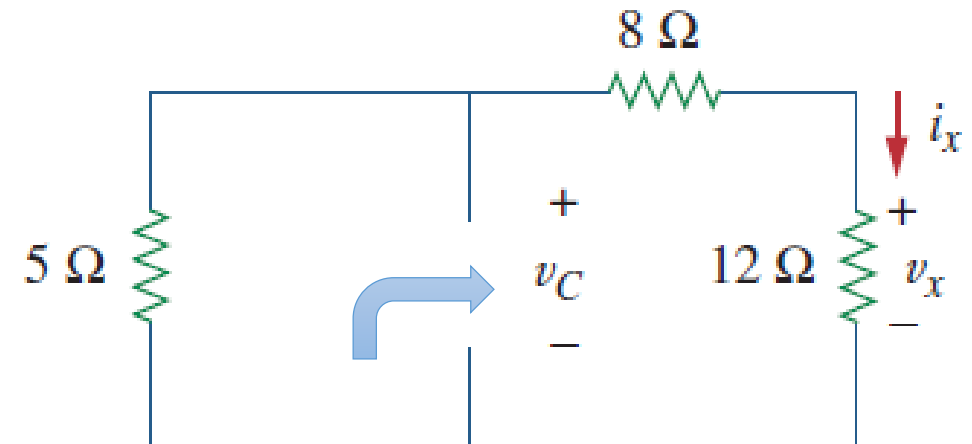
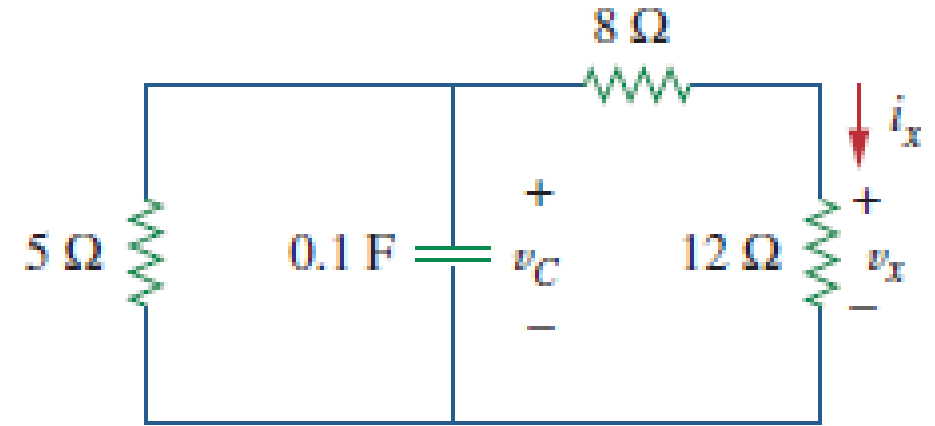
Find v_C , v_x , and i_x for $t > 0$. let $v_C(0) = 15$ V.

Solution:

To find time constant:

$$R_{\text{eq}} = \frac{20 \times 5}{20 + 5} = 4 \Omega$$

$$\tau = R_{\text{eq}}C = 4(0.1) = 0.4 \text{ s}$$

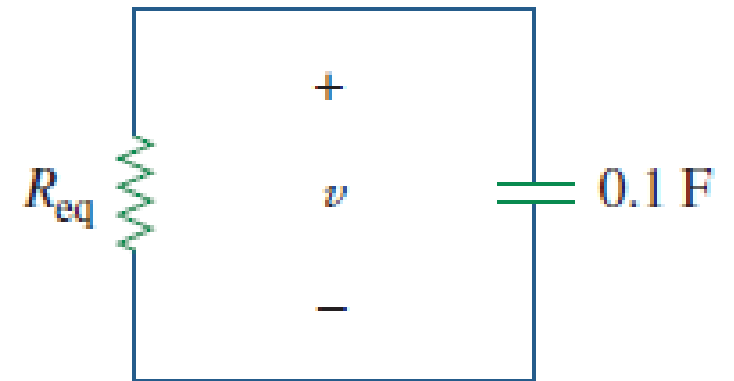


$$v = v(0)e^{-t/\tau} = 15e^{-t/0.4} \text{ V},$$

$$v_C = v = 15e^{-2.5t} \text{ V}$$

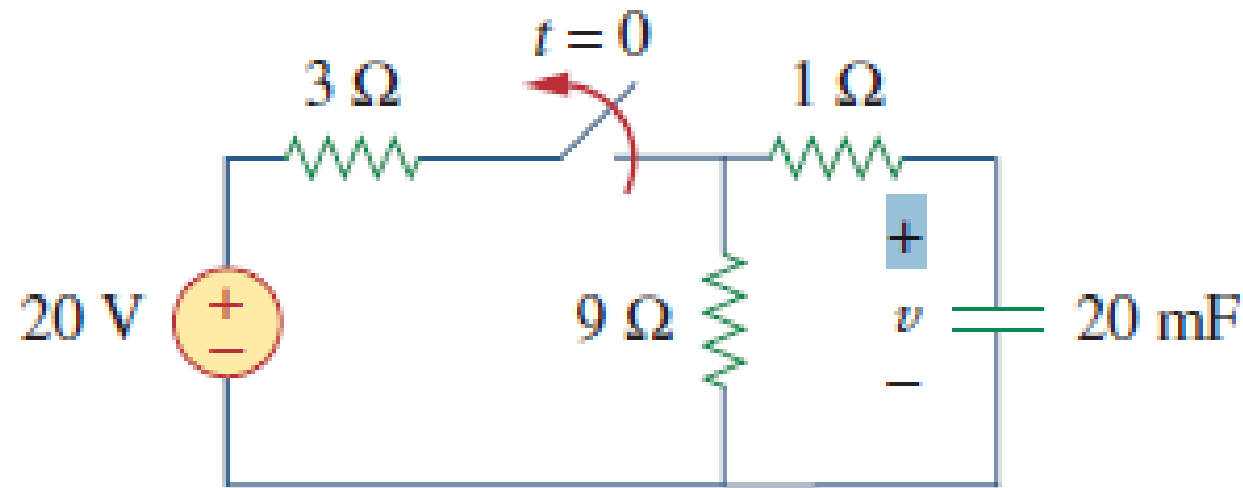
$$v_x = \frac{12}{12 + 8}v = 0.6(15e^{-2.5t}) = 9e^{-2.5t} \text{ V}$$

$$i_x = \frac{v_x}{12} = 0.75e^{-2.5t} \text{ A}$$



Numerical - 2

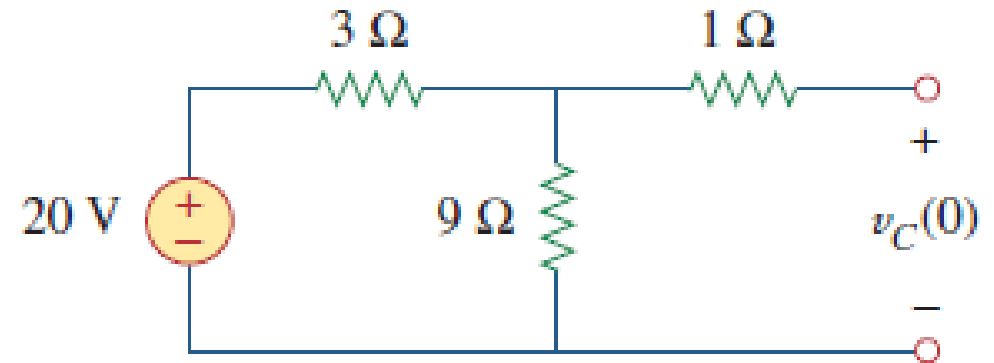
- The switch in the circuit has been closed for a long time, and it is opened at $t = 0$. Find $v(t)$ for $t \geq 0$. Calculate the initial energy stored in the capacitor.



Solution:

For $t < 0$, the switch is closed; the capacitor is an open circuit to dc,

$$v_C(t) = \frac{9}{9 + 3}(20) = 15 \text{ V}, \quad t < 0$$



Since the voltage across a capacitor cannot change instantaneously, the voltage across the capacitor at $t = 0^-$ is the same at $t = 0$, or

$$v_C(0) = V_0 = 15 \text{ V}$$

For $t > 0$, the switch is opened,

$$R_{\text{eq}} = 1 + 9 = 10 \, \Omega$$

The time constant is

$$\tau = R_{\text{eq}}C = 10 \times 20 \times 10^{-3} = 0.2 \, \text{s}$$

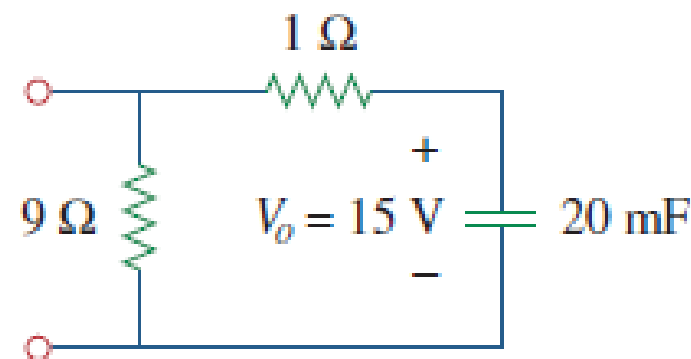
Thus, the voltage across the capacitor for $t \geq 0$ is

$$v(t) = v_C(0)e^{-t/\tau} = 15e^{-t/0.2} \, \text{V}$$

$$v(t) = 15e^{-5t} \, \text{V}$$

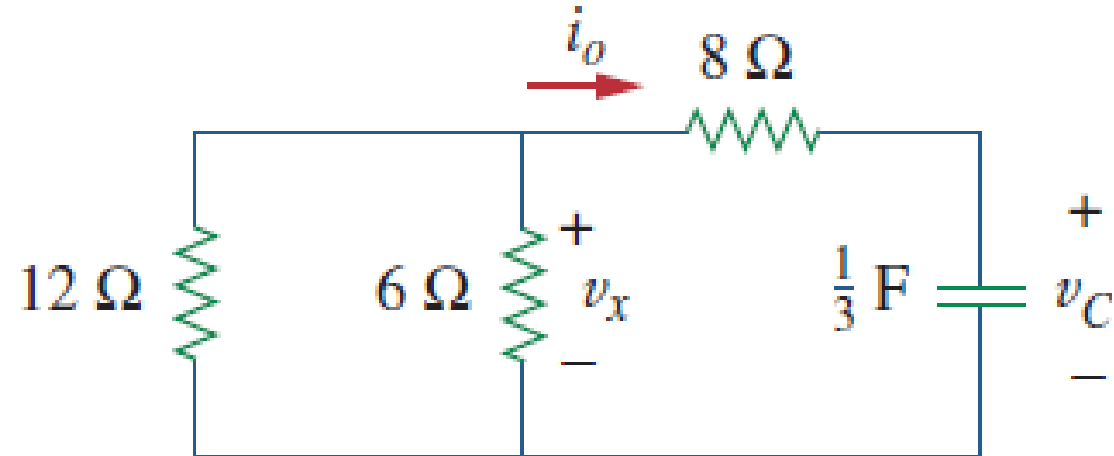
The initial energy stored in the capacitor is

$$w_C(0) = \frac{1}{2}Cv_C^2(0) = \frac{1}{2} \times 20 \times 10^{-3} \times 15^2 = 2.25 \, \text{J}$$



Practice problem- 1

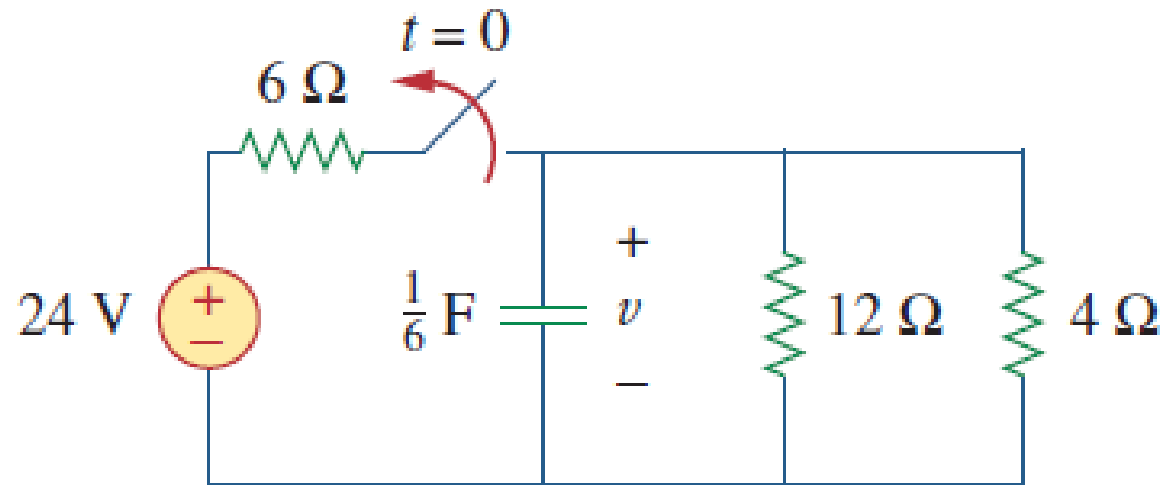
Determine v_C , v_x , and i_o for $t \geq 0$. Let $v_C(0) = 60$ V.



Answer: $60e^{-0.25t}$ V, $20e^{-0.25t}$ V, $-5e^{-0.25t}$ A.

Practice problem- 2

If the switch in Fig. opens at $t = 0$, find $v(t)$ for $t \geq 0$ and $w_C(0)$.



Answer: $8e^{-2t}\text{ V}$, 5.333 J .